

Software Engineering
Term 1 – 2024/2025

Project Proposal Form

- ✓ Group #:4
- ✓ Members of the group:

#	Name	ID	Role
1	Khawla Hani Almoalem	2230003157	Leader
2	Fatimah Moayed Alawa	2230005350	Member
3	Zainab Esaced	2230000664	Member
4	Zahra Aiman Alsadah	2230004418	Member
5	Jood Mohammed Alzayer	2230002118	Member
6	Fatimah Majed Almur	2230004016	Member
7	Remas Nasser Alqahtani	2230002281	Member

- ✓ Title of the Project: **Sanad platform**
- ✓ Introduction of the Project - Problem Statement:

Introduction of the Project: The "Sanad" project is an online platform dedicated to people with disabilities, facilitating their daily lives, both socially and professionally. Our project aims to increase their integration into society by providing a range of services. The platform combines maps to show accessible places, a community for members to communicate with, and other services such as: booking appointments, requesting a volunteer companion for assistance, in addition to providing job opportunities from business owners. Our primary goal is to enable people with disabilities to access their needs with ease and reduce the obstacles to public life.

Problem Statement : Individuals with disabilities frequently encounter major challenges in their daily lives due to poor accessibility, restricted access to affordable services, and insufficient economic inclusion.

Public spaces that are accessible, trustworthy assistance, and specific skills or occupations are hard to achieve.

The current solutions are fragmented: there aren't many social connections, information on how to get around is still scattered, while essential services like technical aids and wheelchairs, and sign language interpretation are difficult to find on a single dashboard.

Autonomy in carrying out daily tasks consequently declines.

- Limited access to public resources and opportunities.

- The lack of hospitable groups promotes loneliness and solitude.
- Obstacles to education and employment that limit full participation in society.

To empower people with disabilities, improve their standard of daily existence, and to promote more inclusion, a comprehensive electronic platform which mixes accessible mapping, community support, critical services, and learning/employment the possibilities is desperately needed.

Objectives (Product Characteristics and Requirements):

1. Making public spaces easier to access:

- Create an interactive map that shows locations (such as cafes, restaurants, hospitals, and universities) that are accessible to individuals with disabilities.

2- Promoting community communication and support:

- Providing a forum for people with disabilities to exchange advice and experiences.
- Showcasing inspiring stories of community successes.

3- Providing digital services and appointments:

- Enabling users to book appointments with sign language interpreters via video.
- offering a "volunteer escort" service for people who need help running particular errands.

4. Improving educational and career prospects:

- Providing training courses online that are accessible to individuals with impairments.
- Linking visitors to hiring websites to provide easily accessible positions.

5. Availability of assistive technology:

- The establishment of an assistive product library, which includes speech devices, screen readers, and smart canes.
- Offering links for these products' purchases or support requests

6- Usability and Accessibility:

- A user interface that is straightforward and easy to use for all disability groups.
- Support for multiple languages to make sure the most users can access it.

Project end user:

- 1- People with disabilities: mobility impairment, hearing impairment, due to their need for services or assistive technologies.
- 2- Families and volunteers: relatives of people with disabilities, or volunteers who assist people with disabilities.
- 3- Service providers: such as sign language interpreters.

4- Employers and educational institutions: entities that provide jobs or training programs.

✓ **Summary of Project Deliverables:**

- 1- Defining Your Project for the Course.
- 2- Submit Project Proposal.
- 3- Submit Project Management Plan (SPMP).
- 4- Submit a status report 1.
- 5- Submit Project Requirements (SRS).
- 6- Submit Project Design (SDS).
- 7- Submit a status report 2.
- 8- Submit Project Test Plan (STS).
- 9- Delivery of the prototype and presentation of the project.

✓ **Project Success Criteria:**

- The development of the platform with an easy-to-use interface that is compatible with the needs of people with disabilities.
- Successful registration of users (people with disabilities, volunteers and employers) and their use of the services provided.
- Providing basic services: (displaying maps of prepared places, the possibility of requesting facilities, booking appointments and publishing job opportunities).
- System stability and free of critical errors during the experiment.
- Achieve initial satisfaction from an experimental user group (e.g., via a survey or live trial).
- Delivery of the project on time according to the plan.

Solution Methodology:

1. Planning

- Gather requirements from all stakeholders (people with disabilities, volunteers, service providers, employers).
- Analyze requirements and transform them into clear platform specifications.
- Develop a comprehensive project plan that includes tasks, timeline, resources, and budget.

2. Design

- Create a user-friendly interface that complies with accessibility standards "WCAG".
- Define the system architecture to illustrate how different modules interact.
- Design database schema and entity relationships.
- Set security measures (data encryption, role-based access control).

3. Implementation

Develop modules independently :

- 1-Map of Accessible Places
- Inspiring Stories and Community Forum
- 3-Volunteer services and appointments
- 4-Section on Education and Employment
- 5-The Library of Assistive Technology.

4 .Testing

- 1-Unit Testing : Verify the operation of every module.
- 2-Integration Testing : Verify that modules work together properly
- 3-User Acceptance Testing (UAT) : Get input from a pilot group of disabled users
- 4-Performance and Security Testing : Verify dependability and safeguard data

5. Deployment

- 1-Use backup and disaster recovery plans when deploying on an AWS or Azure cloud server
- 2-Turn on tools for ongoing uptime and error monitoring

6. Maintenance

- 1-Update the system and fix bugs on a regular basis
- Based on user input enhance accessibility and add new features
- 3-Maintain the database and library of assistive products current.

✓ Signatures of group members

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Date: 25.9.25

----- For instructor use only -----
 Approved _____ Not approved _____ Date:
 Comments: